Prescribed-Time Tracking Control for Nonlinear MASs With Discrete Reference Signals: A Self-Regulating Control Gains Design Method

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***Abstract*—This article addresses the problem of prescribed-time fault-tolerant tracking control for a class of nonlinear multiagent systems (MASs) subject to parameter uncertainties and external disturbances. To improve tracking precision, the trajectory reconstruction approach based on cubic spline interpolation is proposed, which effectively reconstructs the discrete reference signals. Then, a class of prescribed-time regulators is meticulously designed to formulate the fault-tolerant tracking controller, ensuring that the outputs of the controlled system converge to the reconstructed trajectory with arbitrary accuracy within the prescribed tracking time. Finally, stability analyses and a simulation example are presented to demonstrate the effectiveness of the proposed prescribed-time fault-tolerant tracking control strategy, validating its theoretical significance and practical applicability in engineering systems.**

***Index Terms*—Discrete reference trajectory, nonlinear multiagent systems (MASs), prescribed-time tracking control, self-regulating control gains.**

I. Introduction

MULTIAGENT systems (MASs) have garnered considerable research attention in recent decades due to their capacity to address complex tasks through distributed cooperation [1], [2], [3]. The transition from centralized to decentralized control frameworks has been pivotal in advancing various domains, such as robotics, autonomous vehicles, and smart grids, enabling more scalable, flexible, and robust system behaviors [4], [5], [6]. A foundational challenge in MASs pertains to consensus tracking, wherein all agents asymptotically converge to a desired trajectory, guided by a leader [7], [8], [9]. Therefore, ensuring the robustness and efficiency of consensus protocols under various environmental conditions and agent dynamics remains a critical area of research. Under this background, the prescribed-time control strategy has been developed to enhance the control efficiency [10], [11], [12]. For example, an adaptive prescribed-time control controller has been proposed to address time-varying uncertainties and time-delay disturbances in the literature [13]. In addition, the prescribed-time control scheme has been devised for nonlinear systems with sensor faults and unknown control directions, ensuring the convergence of system states within a prescribed time. Despite the fact that these advancements address many theoretical challenges, two significant issues remain. First, existing research has primarily focused on simple nonlinear systems, neglecting the parameter uncertainties that might exist within the system. Second, actuator faults in controlled systems persist as a significant concern. Consequently, there is an urgent requirement to investigate prescribed-time control strategies that also account for actuator faults [14].

In fact, fault-tolerant control plays a crucial role in ensuring the reliability and stability of control systems in the presence of actuator or sensor faults [15]. In MASs, where agents collaborate to achieve a common objective, actuator faults can cause performance degradation or even system failure [16]. The implementation of fault-tolerant control strategies is imperative for enhancing system robustness, thereby enabling the system to maintain effective operation despite such faults. Recent advancements in fault-tolerant control have primarily centered on the development of mechanisms for fault detection, isolation, and reconfiguration, which are critical to ensure system stability under disturbances. For instance, a fault-tolerant consensus protocol for MASs was proposed in the literature [17], providing a solution to achieve consensus even in the presence of communication delays and actuator faults. Additionally, the research in the literature [18] explored the use of adaptive control techniques to handle actuator faults in nonlinear systems, demonstrating that robust fault-tolerant strategies can significantly improve overall system performance.

The dynamic control gain has been introduced to substantially improve the control performance of the system, and the core problem is how to design a suitable regulation

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mechanism to maximize the global control performance while ensuring the control inputs are controllable [19]. For example, the dynamic gain tuning control method has been proposed for automatic generation control in the literature [20], which dynamically adjusts PI controller gains to improve frequency regulation and tie-line flow response under variable wind power. In the literature [21], the dynamic gain method has been developed for global regulation of feed-forward nonlinear systems, using a new time-varying function to design controllers with dynamic gains, which enhances convergence speed and achieves global regulation without the need for feed-forward design. Nevertheless, in nonlinear MASs, the reference trajectory is often received in discrete form, or the system may only receive discrete reference signals when the communication link between the continuous reference trajectory and the leader is disrupted by a network attack or malfunction [22], [23], [24]. Therefore, recovering the original trajectory information from these discrete signals and ensuring continuous tracking control are of significant theoretical research value [25], [26], [27].

Inspired by the aforementioned results, this article proposes a robust prescribed-time tracking control strategy based on the self-regulated gain for uncertain nonlinear MASs, and the primary innovations are summarized as follows.

1) Unlike the event-triggered trajectory reconstruction approaches developed in [28], [29], and [30], a type of trajectory reconstruction function is designed to process discrete reference signals and generate continuous reference trajectories, making the control signals continuous and smooth, thereby reducing oscillations in the controlled system and improving its stability.
2) Compared to the existing methods [12], [31], [32], which cannot address potential actuator failures in MASs, we propose a prescribed-time regulator with multiple design parameters to construct fault-tolerant controllers. This ensures that all agents of the uncertain nonlinear MASs achieve the tracking control objective with arbitrary accuracy within the prescribed tracking time.
3) Based on the error-feedback mechanism, the self-regulating control gain is designed to construct intermediate virtual controllers and fault-tolerant tracking controllers.

In comparison with the controllers using fixed gains [33], [34], [35], our proposed dynamic gains are automatically adjusted based on system errors, maximizing the global control performance of the controlled system while avoiding transient overloads.

II. Problem Statement and Preliminaries

A. System Model

The nonlinear MASs studied in this article consist of n agents, and the agents are described by the following uncertain n_a-order nonlinear system:

$$\begin{cases} \dot{x}_{i,j} = x_{j+1} + \mu_{i,j}\left(\bar{x}_{i,j}\right)^T \theta_{i,j} + d_{i,j}(t) \\ \dot{x}_{i,n_a} = u_i + \mu_{i,n_a}\left(\bar{x}_{i,n_a}\right)^T \theta_{i,n_a} + d_{i,n_a}(t) \\ y_i = x_{i,1}, i = 1,2,\ldots,n, \quad j = 1,2,\ldots,n_a - 1 \end{cases} \tag{1}$$

where $x_{i,k} \in \mathbb{R}$, $\bar{x}_{i,k} = [x_{i,1}, x_{i,2}, \ldots, x_{i,k}]^T \in \mathbb{R}^j$, $\mu_{i,j}(\bar{x}_{i,j}) : \mathbb{R}^j \to \mathbb{R}^{p_i}$, and $u_i \in \mathbb{R}$ are the system states, state vectors, known function vectors, and the control input of the ith agent for $k = 1, 2, \ldots, n_a$, respectively. $\theta_{i,k}$ and $d_{i,k}$ represent the unknown constant vectors and external disturbances, respectively. y_i is the output of the ith agent, $\bar{y}_r$ represents the set of discrete reference signals, and y_r denotes the reconstructed trajectory of the discrete reference signals, which will be estimated by the agents through the adaptive variables $\hat{y}_i$. This means that the estimation errors are $\tilde{y}_i = y_r - \hat{y}_i$, and the tracking errors can be defined as $z_{i,1} = x_{i,1} - \hat{y}_i$. Further, we define that for $t \in [t_0, t_0 + T]$, the reference signal $\bar{y}_r$ is composed of the following set:

$$\begin{cases} \bar{y}_r(t_0) = s(t_0) \\ \bar{y}_r(t_j) = \left\{ s(t_j) \mid t_{j-1} < t_j < t_0 + T, j = 1,2,\ldots,k \right\} \\ \bar{y}_r(t_{k+1}) = s(t_0 + T) \end{cases} \tag{2}$$

where $s(t)$ represents the original reference trajectory, which could not be fully transmitted due to a communication link failure, and k is a positive constant.

Suppose the control inputs u_i will be transformed into the following form in the presence of actuator faults:

$$u_i(t) = \varphi_i(t)\upsilon_i(t) + \bar{u}_i(t) \tag{3}$$

where $\upsilon_i(t)$ denote the designed control inputs. Further, assume $\varphi_i(t)$ and $\bar{u}_i(t)$ satisfy the following properties:

$$\varphi_m \le \varphi_i(t) \le \varphi_M, \qquad |\bar{u}_i(t)| \le \bar{u}_M \tag{4}$$

where φ_m, φ_M, and $\bar{u}_M$ are unknown positive constants. It is important to note that when actuator faults are absent, the control input is equal to the desired input, which means $\varphi_i(t) = 1$ and $\bar{u}_i(t) = 0$.

Definition 1: If for any positive constant $\ell > 0$, $b > |z_{i,1}(0)|$, and $T_f > 0$, the tracking errors $z_{i,1}$ satisfy

$$\begin{aligned} |z_{i,1}(t)| &< \ell, \quad t \ge T_f \\ |z_{i,1}(t)| &\le b, \quad 0 \le t < T_f \end{aligned} \tag{5}$$

then the nonlinear MASs (1) are prescribed-time tracking stable.

1) Control Objective: This article focuses on designing a fault-tolerant tracking control strategy, based on self-regulating gains, for the nonlinear MASs (1) such that the following conditions hold.

1) All the signals in the closed-loop system are bounded, and the tracking errors of the nonlinear MASs (1) are prescribed-time tracking stable.
2) In the presence of actuator faults in the system, it is ensured that the system is stabilized and the control performance is not compromised.

B. Graph Theory

To simplify the representation of the controller design process, we first define the communication topology as below. Assume $\mathcal{B} = \mathrm{diag}(b_1, b_2, \ldots, b_n)$ is a diagonal matrix, where $b_i = 1$ indicates that the ith agent can receive the discrete reference signals $\bar{y}_r$, and $b_i = 0$ otherwise. Consequently, we

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introduce the directed graph $\mathcal{G} \triangleq (\mathcal{V}, \varepsilon)$ to represent the communication link between the agents, where $\mathcal{V} = \{1, 2, \ldots, n\}$ is the set of nodes, with node i corresponding to the ith agent. The set of edges $\varepsilon \subseteq \mathcal{V} \times \mathcal{V}$ represents the directed communication link between distinct agents. An edge (i, j) is in ε if and only if the ith agent can send a message to the jth agent, but not necessarily the other way around. Subsequently, the adjacency matrix $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ is defined as $a_{ii} = 0$, $a_{ij} = 1$ if $(j, i) \in \varepsilon$, and $a_{ij} = 0$ if $(j, i) \notin \varepsilon$. The diagonal matrix $D = \text{diag}(d_1, d_2, \ldots, d_n) \in \mathbb{R}^{n\times n}$ is defined, where $d_i = \sum_{j=1}^{n} a_{ij}$ for $i = 1, 2, \ldots, n$. In the end, we can obtain the Laplacian matrix $L = D - A \in \mathbb{R}^{n\times n}$.

C. Useful Assumptions and Lemma

In order to achieve the control objective, the following assumptions must be made.

Assumption 1: For the external disturbances $d_{i,j}(t)$, there exists an upper bound $d_M > 0$ such that $|d_{i,j}(t)| \le d_M$.

Assumption 2: The directed graph $\mathcal{G}$ is balanced and weakly connected. Additionally, at least one agent can receive the discrete reference signals $\bar{y}_r$, which implies $\sum_{j=1}^{n} b_j > 0$.

Assumption 3: If agent i can receive the messages sent by agent j, such that $a_{ij} = 1$, then agent i can utilize all the state information of agent j for the controller design.

Lemma 1: [28] Define the matrix $\mathcal{Q} = (\mathcal{L} + \mathcal{B}) + (\mathcal{L} + \mathcal{B})^T$. Based on Assumption 3, one can conclude that the matrix $\mathcal{Q}$ is positive definite.

After introducing the preliminaries, we will present the design process of the prescribed-time tracking controller in the next section. Meanwhile, the variables $x_{i,j}$, $\bar{x}_{i,j}$, and t will be omitted from the formulas to simplify the expression.

III. Main Results

The controller design process based on the backstepping framework consists of n_a+1 steps. In these steps, the stabilization of the first n_a subsystems is achieved through constructing virtual controllers, and the fault-tolerant mechanism will be designed in step $n_a + 1$ to eliminate the effects of actuator faults on tracking stability.

A. Prescribed-Time Regulator

In order to guarantee that all agents can effectively track the reference signal within the prescribed time, we establish the following types of PTRs to achieve controller design:

$$\bar{\beta}(t) = \begin{cases} \hbar(t) + \ell, & 0 \le t < T_f \\ \ell, & T_f \le t \end{cases} \tag{6}$$

where ℓ denotes the predefined tracking accuracy and T_f describes the prescribed tracking time, $\hbar(t)$ is the core part of $\bar{\beta}(t)$ and is selected such that the following conditions hold.

1) $\hbar(t)$ is continuous and n_a-order differentiable on $[0, T_f]$.
2) $\hbar(t)$ is monotonically increasing or decreasing on $[0, T_f]$.
3) $\hbar^{(1)}(T_f) = \hbar^{(2)}(T_f) = \ldots = \hbar^{(n_a)}(T_f) = 0$, $\hbar(0) = \gamma - \ell$.

where γ represents the initial error scope. In fact, there are many functions that satisfy the above definition of $T_f(t)$. In this article, we select $(\gamma - \ell)/T_f^{n_a}(T_f - t)^{n_a}$ with a simpler structure. Further, we can obtain the PTR $\beta(t)$ used in the controller design as

$$\beta(t) = \begin{cases} \dfrac{\gamma - \ell}{T_f^{n_a}}(T_f - t)^{n_a} + \ell, & 0 \le t < T_f \\ \ell, & T_f \le t. \end{cases} \tag{7}$$

Remark 1: Several efficient prescribed-time references (PTRs) have been proposed in recent studies under the framework of prescribed-time control [36], [37], [38]. While these PTRs offer promising solutions for achieving time-constrained control, a common issue is that their values at the zero time point tend to infinity, resulting in a singularity in the controllers designed with these PTRs at the initial time. This singularity can lead to undesirable system behaviors and difficulties in controller implementation. To mitigate this challenge, our proposed PTR, denoted as $\beta(t)$, introduces an adjustable parameter γ. By carefully tuning this parameter, we are able to effectively avoid the singularity problem at the initial time.

Define the adaptive parameters and the error transformation as

$$\begin{aligned} \delta_{i,j} &= \left\{ \left\| \theta_{i,j}^T \theta_{i,j} \right\| : i = 1, 2, \ldots, n, j = 1, 2, \ldots, n_a \right\} \\ \varepsilon_{i,1} &= \frac{z_{i,1}}{\beta(t)^2 - z_{i,1}^2}. \end{aligned} \tag{8}$$

It can be further concluded that $\delta_{i,j}$ are unknown positive constants. Therefore, we are going to introduce the adaptive parameters $\hat{\delta}_{i,j}$ to estimate the values of $\delta_{i,j}$. Furthermore, for $|z_{i,1}(0)| < |\beta(0)|$, the errors $\varepsilon_{i,1}(t)$ will remain bounded only if $|z_{i,1}(t)| < |\beta(t)|$ when $t > 0$. Hence, the tracking control objective is to develop a control strategy that ensures $\varepsilon_{i,1}(t)$ is bounded.

B. Dynamic Control Gain Function

Based on the error feedback mechanism, we construct the following dynamic control gain function:

$$H(\kappa) = \frac{e^{\sqrt{1+\kappa^2+c_0}}}{e^{\sqrt{1+\kappa^2+c_0}} - e} \tag{9}$$

where $0 < c_0 < 1$ is a positive design parameter and κ is the error variable. For any value of κ, the function $H(\kappa)$ satisfies the following properties.

1) For any $|\kappa_1| \le |\kappa_2|$, it is clear that $H(\kappa_1) \le H(\kappa_2)$, implying that the value of the dynamic gain function increases as the absolute value of the variable decreases.
2) As κ approaches zero, $\lim_{\kappa\to 0^+} H(\kappa) = (e^{(1+c_0)^{1/2}})/(e^{(1+c_0)^{1/2}} - e)$, implying that choosing a smaller design parameter c_0 results in a larger control gain.
3) When κ tends to positive infinity, the function $H(\kappa)$ approaches a constant value, specifically $\lim_{\kappa\to+\infty} H(\kappa) = 1$.

Remark 2: It is essential to highlight that the dynamic control gain we propose functions as a negative feedback regulator. In contrast to constant gains, as discussed in previous studies [39], [40], [41], our dynamic gains are designed to gradually converge to a constant value when the error

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becomes excessively large, thereby mitigating the risk of excessive control inputs. Simultaneously, when the error is minor, the control gains are dynamically adjusted upward, thereby enhancing overall control performance. Furthermore, the stability analysis section provides a detailed examination of the practical impacts resulting from the implementation of these dynamic control gains.

C. Trajectory Reconstruction Approach

The leaders divide the period into $k-1$ subintervals based on the received discretized trajectory $\bar{y}_r$ and apply the cubic spline interpolation method to reconstruct the desired trajectory. For the subintervals $t \in [t_j, t_{j+1}]$, the reconstructed trajectory is given by the following interpolation function:

$$y_r(t) = \bar{a}_j (t - t_j)^3 + \bar{b}_j (t - t_j)^2 + \bar{c}_j (t - t_j) + \bar{d}_j \tag{10}$$

where $\bar{a}_j$, $\bar{b}_j$, $\bar{c}_j$, and $\bar{d}_j$ are parameters to be determined by the following system of equations:

$$\begin{cases} y_r(t_j) = \bar{y}_r(t_j) \\ y_r(t_{j+1}) = \bar{y}_r(t_{j+1}) \\ y_r(t_j) = y_{r+1}(t_j) \\ \dot{y}_r(t_j) = \dot{y}_{r+1}(t_j) \\ \ddot{y}_r(t_j) = \ddot{y}_{r+1}(t_j), \quad j = 1, 2, \ldots, k. \end{cases} \tag{11}$$

In order to ensure that a unique solution to the system of equations (11) exists, the following boundary conditions are further provided:

$$\begin{cases} \dot{y}_j(t_0) = 0 \\ \ddot{y}_j(t_k) = 0. \end{cases} \tag{12}$$

Now, we are going to prove that the parameters $\bar{a}_j$, $\bar{b}_j$, $\bar{c}_j$ and $\bar{d}_j$ determined by (11) and (12) exist and unique.

Lemma 2: For the interpolating function (10), the parameters $\bar{a}_j$, $\bar{b}_j$, $\bar{c}_j$ and $\bar{d}_j$ exist and unique under the given conditions (11) and (12).

Proof: First, (11) and (12) can be expressed as

$$A\boldsymbol{c} = \boldsymbol{b} \tag{13}$$

where A is a $(k-1) \times (k-1)$ tridiagonal matrix, $\boldsymbol{c}$ is a vector containing the second-order derivatives of the individual nodes, and $\boldsymbol{b}$ is a vector of constants representing the constant terms obtained by the interpolation and continuity conditions. Further, we can deduce that the specific form of the matrix A is as follows:

$$A = \begin{pmatrix} 2(h_1 + h_2) & h_2 & \cdots & 0 \\ h_2 & 2(h_2 + h_3) & \cdots & 0 \\ 0 & h_3 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 2(h_{k-1} + h_k) \end{pmatrix}$$

where $h_j = t_{j+1} - t_j$ represent the length of the subinterval. Assuming that $t_{j+1} \neq t_j$, it follows that the matrix A is non-singular, and that all of its diagonal elements are strictly positive. This condition guarantees that the determinant of A is non-zero, thereby ensuring the existence of a unique solution. The proof is concluded.

Remark 3: Compared to the event-triggered communication, the reconstructed trajectory generated by our proposed approach is $\mathcal{C}^2$-continuous, which not only reduces system oscillations and conserves communication resources, but also smooths all the control signals in the closed-loop system, thus improving the control performance. Comparison tests are designed in the simulation to verify its effectiveness.

D. Prescribed-Time Tracking Controller

Step 1. Define the Lyapunov function as

$$V_{i,1} = \frac{1}{2}\varepsilon_{i,1}^2. \tag{14}$$

Subsequently, it's easy to conclude

$$\begin{aligned} \dot{V}_{i,1} &= \varepsilon_{i,1}\left[\frac{\dot{z}_{i,1}}{\beta^2 - z_{i,1}^2} + \frac{2z_{i,1}^2\dot{z}_{i,1}}{(\beta^2 - z_{i,1}^2)^2} - \frac{2z_{i,1}\beta\dot{\beta}}{(\beta^2 - z_{i,1}^2)^2}\right] \\ &= \varepsilon_{i,1}\varrho_i\dot{z}_{i,1} + \varepsilon_{i,1}\varpi_i \end{aligned} \tag{15}$$

where $\varrho_i = \dot{z}_{i,1}[1/(\beta^2 - z_{i,1}^2) + 2z_{i,1}^2/(\beta^2 - z_{i,1}^2)^2]$, and $\varpi_i = -2z_{i,1}\beta\dot{\beta}/(\beta^2 - z_{i,1}^2)^2$.

By calculating the derivative of the tracking error $z_{i,1}$, we can obtain

$$\dot{z}_{i,1} = x_{i,2} + \mu_{i,1}^T\theta_{i,1} + d_{i,1} - \dot{\hat{y}}_i. \tag{16}$$

Substituting formula (16) into (15), one gives

$$\dot{V}_{i,1} = \varepsilon_{i,1}\varrho_i\left[x_{i,2} + \mu_{i,1}^T\theta_{i,1} + d_{i,1} - \dot{\hat{y}}_i\right] + \varepsilon_{i,1}\varpi_i. \tag{17}$$

The coordinate transformation of the virtual controller is defined as $\varepsilon_{i,2} = x_{i,2} - \alpha_{i,1}$. Consequently, substituting $\varepsilon_{i,2}$ into formula (17), one yields

$$\begin{aligned} \dot{V}_{i,1} = \varepsilon_{i,1}\varrho_i\left[\alpha_{i,1} + \mu_{i,1}^T\theta_{i,1} + d_{i,1} - \dot{\hat{y}}_i\right] \\ + \varepsilon_{i,1}\varepsilon_{i,2}\varrho_i + \varepsilon_{i,1}\varpi_i. \end{aligned} \tag{18}$$

Utilizing Young's inequality, we can further deduce

$$\begin{aligned} \varepsilon_{i,1}\varrho_i\mu_{i,1}^T\theta_{i,1} &\leq \varepsilon_{i,1}^2\varrho_i^2\mu_{i,1}^T\mu_{i,1}\delta_{i,1}p_{i,1} + \frac{1}{4p_{i,1}} \\ \varepsilon_{i,1}\varrho_i d_{i,1} &\leq \varepsilon_{i,1}^2\varrho_i^2 q_{i,1} + \frac{b_M}{4q_{i,1}} \end{aligned} \tag{19}$$

where $p_{i,1}$ and $q_{i,1}$ are positive design parameters. Combining formula (18) with (19), one can conclude

$$\begin{aligned} \dot{V}_{i,1} \leq \varepsilon_{i,1}\varrho_i\left[\alpha_{i,1} + \varepsilon_{i,1}\varrho_i\mu_{i,1}^T\mu_{i,1}\delta_{i,1}p_{i,1} + \varepsilon_{i,1}\varrho_i q_{i,1} - \dot{\hat{y}}_i\right] \\ + \varepsilon_{i,1}\varepsilon_{i,2}\varrho_i + \varepsilon_{i,1}\varpi_i + \frac{1}{4p_{i,1}} + \frac{b_M}{4q_{i,1}}. \end{aligned} \tag{20}$$

Design the virtual controller $\alpha_{i,1}$ as

$$\begin{aligned} \alpha_{i,1} = -\varepsilon_{i,1}\varrho_i\mu_{i,1}^T\mu_{i,1}\hat{\delta}_{i,1}p_{i,1} - \varepsilon_{i,1}\varrho_i q_{i,1} + \dot{\hat{y}}_i - \varpi_i\frac{1}{\varrho_i} \\ - H(\varepsilon_{i,1})\varepsilon_{i,1}\frac{1}{\varrho_i} + h_i\frac{1}{\varrho_i}, \end{aligned} \tag{21}$$

where h_i represents the compensation function. Then, by substituting $\alpha_{i,1}$ into $\dot{V}_{i,1}$, one gives

$$\begin{aligned} \dot{V}_{i,1} \leq \varepsilon_{i,1}\varepsilon_{i,2}\varrho_i - H(\varepsilon_{i,1})\varepsilon_{i,1}^2 + h_i\varepsilon_{i,1} + \Psi_{i,1} \\ + \varepsilon_{i,1}\varrho_i\mu_{i,1}^T\mu_{i,1}\tilde{\delta}_{i,1}p_{i,1}, \end{aligned} \tag{22}$$

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where $\Psi_{i,1}$ is the unknown positive parameter, which has the following form:

$$\Psi_{i,1} = \frac{1}{4p_{i,1}} + \frac{b_M}{4q_{i,1}}. \tag{23}$$

***Step k* ($2 \le k \le n_a - 1$).** Similar to the previous step, the Lyapunov function is chosen to be

$$V_{i,k} = V_{i,k-1} + \frac{1}{2}\varepsilon_{i,k}^2. \tag{24}$$

Subsequently, we can derive the derivative of $V_{i,k}$ as

$$\dot{V}_{i,k} = \dot{V}_{i,k-1} + \varepsilon_{i,k}\left[x_{i,k+1} + \mu_{i,k}^T\theta_{i,k} + d_{i,k} - \dot{\alpha}_{i,k-1}\right]. \tag{25}$$

By utilizing Young's inequality, one gives

$$\begin{aligned}\varepsilon_{i,k}\mu_{i,k}^T\theta_{i,k} &\le \varepsilon_{i,k}^2\mu_{i,k}^T\mu_{i,k}\delta_{i,k}p_{i,k} + \frac{1}{4p_{i,k}}\\ \varepsilon_{i,k}d_{i,k} &\le \varepsilon_{i,k}^2 q_{i,k} + \frac{b_M}{4q_{i,k}}\end{aligned} \tag{26}$$

where $p_{i,k}$ and $q_{i,k}$ are positive design parameters. Substituting (26) into (25) while using the coordinate transformation $\varepsilon_{i,k+1} = x_{i,k+1} - \alpha_{i,k}$, we can conclude

$$\begin{aligned}\dot{V}_{i,k} \le{}& \varepsilon_{i,k}\left[\alpha_{i,k} + \varepsilon_{i,k-1}\varrho_{i,k} + \varepsilon_{i,k}\mu_{i,k}^T\mu_{i,k}\delta_{i,k}p_{i,k} + \varepsilon_{i,k}q_{i,k}\right.\\ &\left. - \dot{\alpha}_{i,k-1}\right] + \Psi_{i,k-1} + \frac{1}{4p_{i,k}} + \frac{b_M}{4q_{i,k}} + \varepsilon_{i,k}\varepsilon_{i,k+1}\\ &+ \sum_{r=1}^{k-1}\varepsilon_{i,r}\varrho_{i,r}\mu_{i,r}^T\mu_{i,r}\tilde{\delta}_{i,r}p_{i,r} + h_i\varepsilon_{i,1}\\ &- \sum_{r=1}^{k-1}H\left(\varepsilon_{i,r}\right)\varepsilon_{i,r}^2\end{aligned} \tag{27}$$

where $\varrho_{i,1} = \varrho_i$, $\varrho_{i,r} = 1$ for $r = 2, 3, \ldots, n_a$. The derivative of the intermediate virtual controller $\alpha_{i,k-1}$ is given as

$$\begin{aligned}\dot{\alpha}_{i,k-1} ={}& \chi_{i,k} + \sum_{j=1}^{n}\sum_{k_i=1}^{k-1} a_{ij}\frac{\partial\alpha_{i,k-1}}{\partial x_{j,k_i}}\left(\mu_{j,k_i}^T\theta_{j,k_i} + d_{j,k_i}\right)\\ &+ \sum_{k_i=1}^{k-1}\frac{\partial\alpha_{i,k-1}}{\partial x_{i,k_i}}\left(\mu_{i,k_i}^T\theta_{i,k_i} + d_{i,k_i}\right).\end{aligned} \tag{28}$$

The variable $\chi_{i,k}$ is a known variable of the following form:

$$\begin{aligned}\chi_{i,k} ={}& \sum_{j=1}^{n}\sum_{k_i=1}^{k-1} a_{ij}\frac{\partial\alpha_{i,k-1}}{\partial x_{j,k_i}}x_{j,k_i+1} + \sum_{k_i=1}^{k-1} b_i\frac{\partial\alpha_{i,k-1}}{\partial y_r^{(k_i-1)}}y_r^{k_i}\\ &+ \frac{\partial\alpha_{i,k-1}}{\partial\hat{y}_i}\dot{\hat{y}}_i + \sum_{k_i=1}^{k-1}\frac{\partial\alpha_{i,k-1}}{\partial\hat{\delta}_{i,k_i}}\dot{\hat{\delta}}_{i,k_i}\\ &+ \sum_{k_i=1}^{k-1}\frac{\partial\alpha_{i,k-1}}{\partial x_{i,k_i}}x_{i,k_i+1}.\end{aligned} \tag{29}$$

By utilizing Young's inequality, it can be deduced that

$$\begin{aligned}-\frac{\partial\alpha_{i,k-1}}{\partial x_{j,k_i}}\mu_{j,k_i}^T\theta_{j,k_i}\varepsilon_{i,k} &\le \frac{\bar{\delta}}{4\bar{p}}\\ &+ \left(\frac{\partial\alpha_{i,k-1}}{\partial x_{j,k_i}}\right)^2\mu_{j,k_i}^T\mu_{j,k_i}\varepsilon_{i,k}^2\bar{p}\\ -\frac{\partial\alpha_{i,k-1}}{\partial x_{j,k_i}}d_{j,k_i}\varepsilon_{i,k} &\le \left(\frac{\partial\alpha_{i,k-1}}{\partial x_{j,k_i}}\right)^2\varepsilon_{i,k}^2\bar{p} + \frac{b_M}{4\bar{p}}\end{aligned} \tag{30}$$

where $\bar{\delta} = \max\{\delta_{i,j}\}$, $j = 1, 2, \ldots, n_i$. Now, constructing the virtual controller $\alpha_{i,k}$ as

$$\begin{aligned}\alpha_{i,k} ={}& -H\left(\varepsilon_{i,k}\right)\varepsilon_{i,k} - \varepsilon_{i,k}\mu_{i,k}^T\mu_{i,k}\hat{\delta}_{i,k}p_{i,k} - \varepsilon_{i,k}q_{i,k}\\ &+ \chi_{i,k} - \psi_{i,k}\varepsilon_{i,k} - \varepsilon_{i,k-1}\varrho_{i,k},\end{aligned} \tag{31}$$

where $\psi_{i,k}$ is an unknown variable of the following form:

$$\begin{aligned}\psi_{i,k} ={}& \sum_{j=1}^{n}\sum_{k_i=1}^{k-1} a_{ij}\left[\left(\frac{\partial\alpha_{i,k-1}}{\partial x_{j,k_i}}\right)^2\mu_{j,k_i}^T\mu_{j,k_i}\bar{p}\right.\\ &\left. + \left(\frac{\partial\alpha_{i,k-1}}{\partial x_{j,k_i}}\right)^2\bar{p}\right] + \sum_{k_i=1}^{k-1}\left(\frac{\partial\alpha_{i,k-1}}{\partial x_{i,k_i}}\right)^2\bar{p}\\ &+ \sum_{k_i=1}^{k-1}\left(\frac{\partial\alpha_{i,k-1}}{\partial x_{i,k_i}}\right)^2\mu_{i,k_i}^T\mu_{i,k_i}\bar{p}.\end{aligned} \tag{32}$$

Then, by combining $\alpha_{i,k}$ with (27), one can derive

$$\begin{aligned}\dot{V}_{i,k} \le{}& -\sum_{r=1}^{k}H\left(\varepsilon_{i,r}\right)\varepsilon_{i,r}^2 + \sum_{r=1}^{k}\varepsilon_{i,r}\varrho_{i,r}\mu_{i,r}^T\mu_{i,r}\tilde{\delta}_{i,r}p_{i,r}\\ &+ h_i\varepsilon_{i,1} + \varepsilon_{i,k}\varepsilon_{i,k+1}\end{aligned} \tag{33}$$

where $\Psi_{i,k}$ is the unknown positive parameter, which has the following form:

$$\begin{aligned}\Psi_{i,k} ={}& \Psi_{i,k-1} + \left[\sum_{j=1}^{n}a_{ij} + 1\right]\frac{\bar{\delta} + b_M}{4\bar{p}}\\ &+ \frac{1}{4p_{i,k}} + \frac{b_M}{4q_{i,k}}.\end{aligned} \tag{34}$$

Step n_a. In this step, we are going to construct the control input u_i. First, we select the following Lyapunov function:

$$V_{i,n_a} = V_{i,n_a-1} + \frac{1}{2\varphi_m}\varepsilon_{i,n_a}^2. \tag{35}$$

Consequently, the derivative of V_{i,n_a} is derived as

$$\begin{aligned}\dot{V}_{i,n_a} = \dot{V}_{i,n_a-1} + \frac{1}{\varphi_m}\varepsilon_{i,n_a}\left[\varphi_i\upsilon_i + \bar{u}_i + \mu_{i,n_a}^T\theta_{i,n_a}\right.\\ \left. + d_{i,n_a} - \dot{\alpha}_{i,n_a-1}\right].\end{aligned} \tag{36}$$

Deriving the intermediate virtual controller yields

$$\begin{aligned}\dot{\alpha}_{i,n_a-1} ={}& \chi_{i,n_a} + \sum_{j=1}^{n}\sum_{k_i=1}^{n_a-1} a_{ij}\frac{\partial\alpha_{i,k-1}}{\partial x_{j,k_i}}\left(\mu_{j,k_i}^T\theta_{j,k_i} + d_{j,k_i}\right)\\ &+ \sum_{k_i=1}^{n_a-1}\frac{\partial\alpha_{i,n_a-1}}{\partial x_{i,k_i}}\left(\mu_{i,k_i}^T\theta_{i,k_i} + d_{i,k_i}\right)\end{aligned} \tag{37}$$

where the variable χ_{i,n_a} is a known variable of the following form:

$$\begin{aligned}\chi_{i,n_a} ={}& \sum_{j=1}^{n}\sum_{k_i=1}^{n_a-1} a_{ij}\frac{\partial\alpha_{i,n_a-1}}{\partial x_{j,k_i}}x_{j,k_i+1} + \sum_{k_i=1}^{n_a-1} b_i\frac{\partial\alpha_{i,n_a-1}}{\partial y_r^{(k_i-1)}}y_r^{k_i}\\ &+ \frac{\partial\alpha_{i,n_a-1}}{\partial\hat{y}_i}\dot{\hat{y}}_i + \sum_{k_i=1}^{n_a-1}\frac{\partial\alpha_{i,n_a-1}}{\partial\hat{\delta}_{i,k_i}}\dot{\hat{\delta}}_{i,k_i}\\ &+ \sum_{k_i=1}^{n_a-1}\frac{\partial\alpha_{i,n_a-1}}{\partial x_{i,k_i}}x_{i,k_i+1}.\end{aligned} \tag{38}$$

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Further, the following unknown variable is introduced:

$$\psi_{i,n_a} = \sum_{j=1}^{n}\sum_{k_i=1}^{n_a-1} a_{ij}\left[\left(\frac{\partial \alpha_{i,n_a-1}}{\partial x_{j,k_i}}\right)^2 \mu_{j,k_i}^T \mu_{j,k_i}\bar{p} + \left(\frac{\partial \alpha_{i,n_a-1}}{\partial x_{j,k_i}}\right)^2 \bar{p}\right] + \sum_{k_i=1}^{n_a-1}\left(\frac{\partial \alpha_{i,n_a-1}}{\partial x_{i,k_i}}\right)^2 \bar{p} + \sum_{k_i=1}^{n_a-1}\left(\frac{\partial \alpha_{i,n_a-1}}{\partial x_{i,k_i}}\right)^2 \mu_{i,k_i}^T \mu_{i,k_i}\bar{p}. \tag{39}$$

Similar to formula (30), one can derive

$$-\frac{\partial \alpha_{i,n_a-1}}{\partial x_{j,k_i}\varphi_m}\mu_{j,k_i}^T\theta_{j,k_i}\varepsilon_{i,n_a} \le \left(\frac{\partial \alpha_{i,n_a-1}}{\partial x_{j,k_i}}\right)^2 \mu_{j,k_i}^T\mu_{j,k_i}\varepsilon_{i,k}^2\bar{p} + \frac{\bar{\delta}}{4\varphi_m^2\bar{p}}$$
$$-\frac{\partial \alpha_{i,n_a-1}}{\partial x_{j,k_i}\varphi_m} d_{j,k_i}\varepsilon_{i,n_a} \le \left(\frac{\partial \alpha_{i,n_a-1}}{\partial x_{j,k_i}}\right)^2 \varepsilon_{i,n_a}^2\bar{p} + \frac{b_M}{4\varphi_m^2\bar{p}}. \tag{40}$$

Remark 4: It should be noted that in inequality (40), a distinct scaling approach is employed to mitigate the impact of the uncertainty term $\theta_{j,k}$, thus avoiding the complexity explosion that could arise from the incorporation of neighboring information. Additionally, the introduction of the adaptive mechanism $\hat{\bar{\delta}}_{i,k}$ compensates for the effects of uncertain parameters on the control signal, improving the global control performance.

Then, by using Young's inequality, one can obtain

$$\begin{aligned}
\frac{\varepsilon_{i,n_a}}{\varphi_m}\mu_{i,n_a}^T\theta_{i,n_a} &\le \varepsilon_{i,n_a}^2\mu_{i,n_a}^T\mu_{i,n_a}\delta_{i,n_a}p_{i,n_a} + \frac{1}{4\varphi_m^2 p_{i,n_a}}\\
\frac{1}{\varphi_m}\varepsilon_{i,n_a-1}\varepsilon_{i,n_a} &\le \varepsilon_{i,n_a-1}^2\varepsilon_{i,n_a}^2 c_{i,2} + \frac{1}{4\varphi_m^2 c_{i,2}}\\
\frac{1}{\varphi_m}\varepsilon_{i,n_a}d_{i,n_a} &\le \varepsilon_{i,n_a}^2 q_{i,n_a} + \frac{b_M}{4\varphi_m^2 q_{i,n_a}}\\
\frac{1}{\varphi_m}\varepsilon_{i,n_a}\bar{u}_i &\le \varepsilon_{i,n_a}^2 c_{i,3} + \frac{\bar{u}_M}{4\varphi_m^2 c_{i,3}}\\
-\frac{1}{\varphi_m}\chi_{i,n_a}\varepsilon_{i,n_a} &\le \chi_{i,n_a}^2\varepsilon_{i,n_a}^2 c_{i,1} + \frac{1}{4\varphi_m^2 c_{i,1}}.
\end{aligned} \tag{41}$$

where p_{i,n_a}, q_{i,n_a}, $c_{i,1}$, $c_{i,2}$, and $c_{i,3}$ are positive design parameters. By substituting formulas (37)–(41) into (36), we can conclude

$$\dot{V}_{i,n_a} \le -\sum_{r=1}^{n_a-1} H\left(\varepsilon_{i,r}\right)\varepsilon_{i,r}^2 + \varepsilon_{i,n_a}^2\left[\mu_{i,n_a}^T\mu_{i,n_a}\delta_{i,n_a}p_{i,n_a} + \frac{\varphi_i}{\varphi_m\varepsilon_{i,n_a}}\upsilon_i + q_{i,n_a} + \varepsilon_{i,n_a-1}^2 c_{i,2} + \chi_{i,n_a}^2 c_{i,1} + \psi_{i,n_a} + c_{i,3}\right] + h_i\varepsilon_{i,1} + \Psi_{i,n_a-1} - \sum_{r=1}^{n_a-1}\varepsilon_{i,r}\varrho_{i,r}\mu_{i,r}^T\mu_{i,r}\tilde{\bar{\delta}}_{i,r}p_{i,r} \tag{42}$$

where Ψ_i is the unknown positive parameter, which has the following form:

$$\Psi_i = \Psi_{i,n_a-1} + \left[\sum_{j=1}^{n} a_{ij} + 1\right](n_a-1)\frac{\bar{\delta}+b_M}{4\bar{p}\varphi_m^2} + \frac{1}{4\varphi_m^2 c_{i,1}} + \frac{1}{4\varphi_m^2 c_{i,2}} + \frac{b_M}{4\varphi_m^2 q_{i,n_a}} + \frac{\bar{u}_M}{4\varphi_m^2 c_{i,3}} + \frac{1}{4\varphi_m^2 p_{i,n_a}}. \tag{43}$$

Constructing the desired control inputs υ_i as

$$\begin{aligned}
\upsilon_i = &-\varepsilon_{i,n_a}c_{i,3} - \varepsilon_{i,n_a}\mu_{i,n_a}^T\mu_{i,n_a}\hat{\bar{\delta}}_{i,n_a}p_{i,n_a} - \varepsilon_{i,n_a}q_{i,n_a}\\
&- \varepsilon_{i,n_a-1}^2\varepsilon_{i,n_a}c_{i,2} - \varepsilon_{i,n_a}\chi_{i,n_a}^2 c_{i,1}\\
&- \psi_{i,k}\varepsilon_{i,n_a} - H\left(\varepsilon_{i,n_a}\right)\varepsilon_{i,n_a}.
\end{aligned} \tag{44}$$

Now, combining formula (42) with (44), one gives

$$\dot{V}_{i,n_a} \le -\sum_{r=1}^{n_a} H\left(\varepsilon_{i,r}\right)\varepsilon_{i,r}^2 + \sum_{r=1}^{n_a}\varepsilon_{i,r}\varrho_{i,r}\mu_{i,r}^T\mu_{i,r}\tilde{\bar{\delta}}_{i,r}p_{i,r} + \Psi_i + h_i\varepsilon_{i,1}. \tag{45}$$

Step n_a + 1. In order to eliminate the nonlinear uncertainty term $\sum_{r=1}^{n_a}\varepsilon_{i,r}\varrho_{i,r}\mu_{i,r}^T\mu_{i,r}\tilde{\bar{\delta}}_{i,r}p_{i,r}$, we will design the plausible adaptive laws $\dot{\hat{\bar{\delta}}}_{i,r}$, $i = 1,2,\ldots,n$ and $r = 1,2,\ldots,n_a$. Meanwhile, we are going to prove the boundedness of the closed-loop system by introducing the estimate error $\tilde{y}_i = y_r - \hat{y}_i$ into the Lyapunov function.

Construct the positive definite function W_i as

$$W_i = V_{i,n_a} + \frac{1}{2}\sum_{r=1}^{n_a}\tilde{\bar{\delta}}_{i,r}^2 + \frac{1}{2}\tilde{y}_i^2. \tag{46}$$

Subsequently, the derivate of W_i is derived as

$$\dot{W}_i = \dot{V}_{i,n_a} - \sum_{r=1}^{n_a}\tilde{\bar{\delta}}_{i,r}\dot{\hat{\bar{\delta}}}_{i,r} - \tilde{y}_i\dot{\hat{y}}_i. \tag{47}$$

Now, design the adaptive laws $\dot{\hat{\bar{\delta}}}_{i,r}$ as follows:

$$\dot{\hat{\bar{\delta}}}_{i,r} = \varepsilon_{i,r}\varrho_{i,r}\mu_{i,r}^T\mu_{i,r}p_{i,r} - \bar{p}_{i,r}\hat{\bar{\delta}}_{i,r} \tag{48}$$

where $i = 1,2,\ldots,n$, $r = 1,2,\ldots,n_a$, and $\bar{p}_{i,r}$ are positive design parameters. Then, combining the adaptive laws $\dot{\hat{\bar{\delta}}}_{i,r}$ with the derivate of W_i, we can conclude

$$\dot{W}_i \le -\sum_{r=1}^{n_a} H\left(\varepsilon_{i,r}\right)\varepsilon_{i,r}^2 + \sum_{r=1}^{n_a}\bar{p}_{i,r}\hat{\bar{\delta}}_{i,r}\tilde{\bar{\delta}}_{i,r} - \tilde{y}_i\dot{\hat{y}}_i + \Psi_i + h_i\varepsilon_{i,1}. \tag{49}$$

By observing formula (49), we can obtain that the nonlinear terms $\tilde{y}_i\dot{\hat{y}}_i$ may affect the stability of the closed-loop system. Therefore, we utilize the compensation functions h_i to address this issue. Constructing h_i as follows:

$$h_i = -g_i\left(\beta^2 - z_{i,1}^2\right)\left[\sum_{j=1}^{n} a_{ij}\left(y_i - y_j\right) + b_i\left(y_i - y_r\right)\right] \tag{50}$$

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where g_i is a positive design parameter. Then, we can further conclude

$$h_i\varepsilon_{i,1} = -g_i(y_i - y_r)\left[\sum_{j=1}^{n} a_{ij}(y_i - y_j) + b_i(y_i - y_r)\right] - g_i\tilde{y}_i\left[\sum_{j=1}^{n} a_{ij}(y_i - y_j) + b_i(y_i - y_r)\right]. \tag{51}$$

By substituting formula (51) into (49), one yields

$$\dot{W}_i \le -g_i\tilde{y}_i\left[\sum_{j=1}^{n} a_{ij}(y_i - y_j) + b_i(y_i - y_r) + \frac{1}{g_i}\dot{\hat{y}}_i\right] - g_i(y_i - y_r)\left[\sum_{j=1}^{n} a_{ij}(y_i - y_j) + b_i(y_i - y_r)\right] - \sum_{r=1}^{n_a} H(\varepsilon_{i,r})\varepsilon_{i,r}^2 + \sum_{j=1}^{n_a}\sum_{r=1}^{n_a}\bar{p}_{i,r}\hat{\delta}_{i,r}\tilde{\delta}_{i,r} + \Psi_i. \tag{52}$$

Now, designing the adaptive laws $\dot{\hat{y}}_i$ as

$$\dot{\hat{y}}_i = -g_i\left[\sum_{r=1}^{n} a_{ij}(y_i - y_j) + b_i(y_i - y_r)\right] - a_i\hat{y}_i \tag{53}$$

where a_i are the positive design parameters. Subsequently, combining $\dot{\hat{y}}_i$ with formula (52), one gives

$$\dot{W}_i \le -g_i(y_i - y_r)\left[\sum_{j=1}^{n} a_{ij}(y_i - y_j) + b_i(y_i - y_r)\right] + \Psi_i - \sum_{r=1}^{n_a} H(\varepsilon_{i,r})\varepsilon_{i,r}^2 + \sum_{r=1}^{n_a}\bar{p}_{i,r}\hat{\delta}_{i,r}\tilde{\delta}_{i,r} + a_i\tilde{y}_i\hat{y}_i. \tag{54}$$

Up to this point, the design of all control signals within the closed-loop system has been completed. In the subsequent analysis, we will examine the stability of the closed-loop system, leveraging the control signals that have been formulated in the preceding steps.

E. Stability Analysis

Theorem 1: Consider the nonlinear MASs (1) under ***Assumptions 1–3*** with actuator faults modeled by (4), there exists a group of control signals specified in formulas (21), (31), (44), (48), (50), and (53) such that the following results hold.

1) All signals in the closed-loop system are bounded, and the tracking errors of the nonlinear MAS (1) realize the prescribed-time tracking stability.
2) In the presence of actuator faults, the system is guaranteed to remain stabilized, ensuring that control performance does not degrade.
3) All the errors in the closed-loop system will converge to a fixed compact set.

Proof: In accordance with the established definition of actuator faults given by (4), it can be deduced that when the ith agent is fault-free, $\varphi_i(t) = 1$ and $\bar{u}_i(t) = 0$. This fault-free status can thus be considered as a special case. Therefore, the present analysis is focused on investigating the stability of the controlled system, specifically in the presence of actuator faults.

1) Initially, the demonstration will be made that all signals in the closed-loop system are bounded, and that the tracking error is prescribed-time tracking stable. Select the Lyapunov function W as follows:

$$W = \sum_{i=1}^{n} W_i. \tag{55}$$

Therefore, it's easy to obtain

$$\dot{W} \le -\sum_{i=1}^{n} g_i(y_i - y_r)\left[\sum_{j=1}^{n} a_{ij}(y_i - y_j) + b_i(y_i - y_r)\right] + \sum_{i=1}^{n}\left[\sum_{r=1}^{n_a}\bar{p}_{i,r}\hat{\delta}_{i,r}\tilde{\delta}_{i,r} - \sum_{r=1}^{n_a} H(\varepsilon_{i,r})\varepsilon_{i,r}^2 + \Psi_i + a_i\tilde{y}_i\hat{y}_i\right]. \tag{56}$$

By replacing the invariant variable with an unknown positive constant $\ell = \sum_{i=1}^{n}\Psi_i$, we can further conclude

$$\dot{W} \le -\sum_{i=1}^{n} g_i(y_i - y_r)\left[\sum_{j=1}^{n} a_{ij}(y_i - y_j) + b_i(y_i - y_r)\right] + \sum_{i=1}^{n}\sum_{r=1}^{n_a}\bar{p}_{i,r}\hat{\delta}_{i,r}\tilde{\delta}_{i,r} - \sum_{i=1}^{n}\sum_{r=1}^{n_a} H(\varepsilon_{i,r})\varepsilon_{i,r}^2 + \sum_{i=1}^{n} a_i\tilde{y}_i\hat{y}_i + \ell. \tag{57}$$

Using Young's inequality, one gives

$$\tilde{\delta}_{i,j}\hat{\delta}_{i,j} \le -\frac{1}{2}\tilde{\delta}_{i,j}^2 + \frac{1}{2}\delta_{i,j}^2,\ \tilde{y}_i\hat{y}_i \le -\frac{1}{2}\tilde{y}_i^2 + \frac{1}{2}y_r^2. \tag{58}$$

Since $g_i > 0$, combining formula (58) with (57), it's easy to verify

$$\dot{W} \le -\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n_a}\bar{p}_{i,j}\tilde{\delta}_{i,j}^2 - \sum_{i=1}^{n}\sum_{r=1}^{n_a} H(\varepsilon_{i,r})\varepsilon_{i,r}^2 - \frac{1}{2}\sum_{i=1}^{n} a_i\tilde{y}_i^2 + \ell + \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n_a}\bar{p}_{i,j}\delta_{i,j}^2 + \frac{1}{2}\sum_{i=1}^{n} a_i y_r^2 \le -aW + \bar{\ell} \tag{59}$$

where a is the minimum design parameters, and $\bar{\ell} = \ell + 1/2\sum_{i=1}^{n}\sum_{j=1}^{n_a}\bar{p}_{i,j}\delta_{i,j}^2 + 1/2\sum_{i=1}^{n} a_i y_r^2$. Taking the integral on both sides of the inequality (59) yields

$$W(t) \le \left(W(0) - \frac{\bar{\ell}}{a}\right)e^{-at} + \frac{\bar{\ell}}{a},\quad t \ge 0 \tag{60}$$

which indicates that for $\forall t \ge 0$, the error variable $\xi = [\varepsilon_{i,j}, \tilde{y}_i, \tilde{\delta}_{i,j}]$, $i = 1, 2, \ldots, n$, $j = 1, 2, \ldots, n_a$ belongs to the compact set

$$\Omega_1 = \left(\xi \,\middle|\, W(\xi(t)) \le W(0) + \frac{\bar{\ell}}{a}\right). \tag{61}$$

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Therefore, we can further deduce that all the errors, including the tracking errors $\varepsilon_{i,1}$, the system errors $\varepsilon_{i,\gamma}$, $\gamma = 2,3,\ldots,n_a$, and the estimation errors $\tilde{y}_i$ and $\tilde{\delta}_{i,j}$ in the closed-loop system are bounded. Since y_r and $\delta_{i,j}$ are bounded, we can obtain that $\hat{y}_i$ and $\hat{\delta}_{i,j}$ are bounded. Further, based on the above formulas, one gives that the virtual controllers $\alpha_{i,k}$ and the control input u_i are bounded, where $i = 1,2,\ldots,n$, $k = 1,2,\ldots,n_a$. Hence, all the closed-loop signals are uniformly bounded. Meanwhile, based on the definitions of the coordinate transformation $\varepsilon_{i,1}$, and the prescribed-time tracking stability described in (5), we can further deduce that the tracking error will attain the prescribed-time tracking stability when all the signals within the closed-loop system are bounded.

2) Now, we are going to prove that all the tracking errors in the closed-loop system will converge to a fixed compact set. Based on formula (59), we can derive the following conclusion:

$$\begin{aligned}\dot{W} &\le -\sum_{i=1}^{n}\sum_{r=1}^{n_a}\frac{e^{\sqrt{1+\varepsilon_{i,r}^2}+c_0}}{e^{\sqrt{1+\varepsilon_{i,r}^2}+c_0}-e}\varepsilon_{i,r}^2+\bar{\ell}\\ &\le -\sum_{i=1}^{n}\sum_{r=1}^{n_a}\frac{e^{\sqrt{1+V_s}+c_0}}{e^{\sqrt{1+V_s}+c_0}-e}\varepsilon_{i,r}^2+\bar{\ell}\\ &\le -\frac{e^{V_s+c_0}}{e^{V_s+c_0}-1}\sum_{i=1}^{n}\sum_{r=1}^{n_a}\varepsilon_{i,r}^2+\bar{\ell}\\ &\le -\frac{e^{V_s+c_0}}{e^{V_s+c_0}-1}V_s+\bar{\ell}\end{aligned} \tag{62}$$

where $V_s = \sum_{i=1}^{n}\sum_{r=1}^{n_a}\varepsilon_{i,r}^2$ denotes the potential energy possessed by the tracking errors. Further, the following nonlinear auxiliary function is constructed:

$$F(V_s) = -\frac{e^{V_s+c_0}}{e^{V_s+c_0}-1}V_s+\bar{\ell}. \tag{63}$$

The derivative of $F(V_s)$ is calculated as follows:

$$\dot{F}(V_s) = \frac{e^{V_s+c_0}}{\left(e^{V_s+c_0}-1\right)^2}V_s - \frac{e^{V_s+c_0}}{e^{V_s+c_0}-1}. \tag{64}$$

It is evident that the derivative of F_s is negative. Consequently, we can infer that V_s will asymptotically converge to a new compact set, described by

$$\Omega_2 = \left\{V_s \left| \bar{c}\frac{e^{V_s+c_0}}{e^{V_s+c_0}-1}V_s \le \bar{\ell}\right.\right\} \tag{65}$$

which implies that the parameter c_0 can be adjusted to drive the error variable of the closed-loop system toward a fixed compact set.

Remark 5: It is important to note that, given the limitations of current mathematical methods, the exact solution to the implicit inequality (65) cannot be derived. However, by applying numerical techniques such as Newton's method, an approximate solution for this compact set can be obtained. The results demonstrate that the compact set Ω_2 is significantly smaller than Ω_1, despite having identical design parameters.

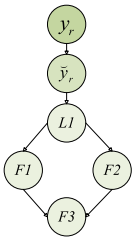


Fig. 1. Communication topology of the controlled MAS.

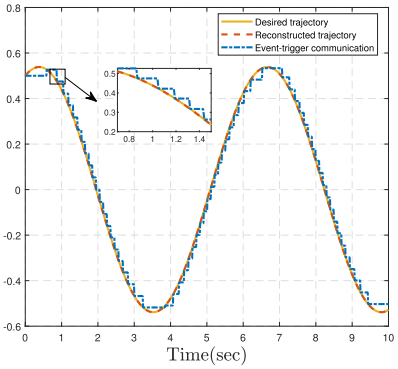
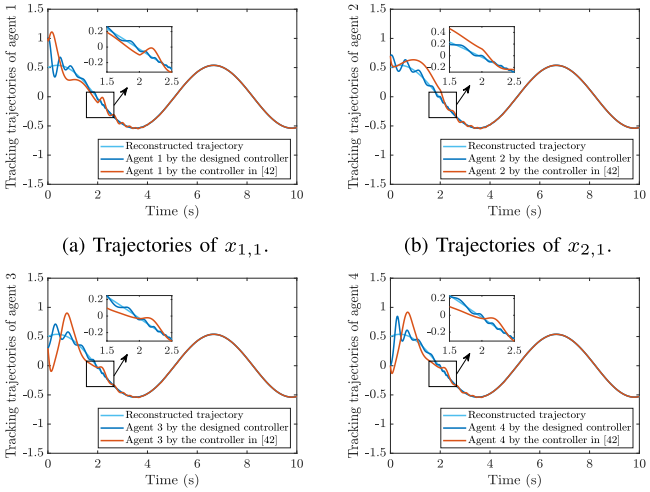


Fig. 2. Reconstruction performance of different algorithms.



(a) Trajectories of $x_{1,1}$. (b) Trajectories of $x_{2,1}$.

(c) Trajectories of $x_{3,1}$. (d) Trajectories of $x_{4,1}$.

Fig. 3. Tracking trajectories of the controlled MAS. (a) Trajectories of $x_{1,1}$. (b) Trajectories of $x_{2,1}$. (c) Trajectories of $x_{3,1}$. (d) Trajectories of $x_{4,1}$.

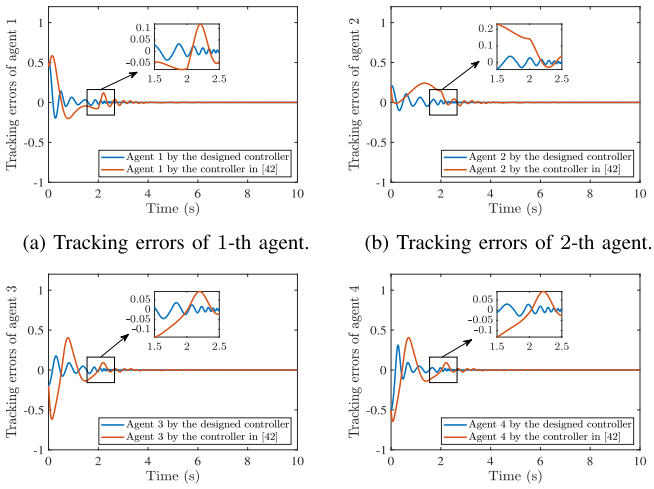
IV. Simulation Example

This section provides a simulation example to examinethe effectiveness of the proposed control strategy. Consider the following uncertain nonlinear MAS:

$$\begin{cases}\dot{x}_{i,1} = x_{i,2} + \dfrac{1}{4}x_{i,1}^3 + e^{-x_{i,1}} + \sin^2 x_{i,1} + \dfrac{1}{10}\cos^2 t\\ \dot{x}_{i,2} = u_i + \dfrac{3}{10}x_{i,1}x_{i,2} + \dfrac{1}{5}\sin x_{i,2}^3 + \dfrac{1}{10}\sin^2 t\\ y_i = x_{i,1}\end{cases} \tag{66}$$

where $\mu_{i,1} = [x_{i,1}^3, e^{-x_{i,1}}, \sin^2 x_{i,1}]$, $\theta_{i,1} = [0.25,1,1]^T$, $\mu_{i,2} = [x_{i,1}x_{i,2}, \sin x_{i,2}^3]$, $\theta_{i,2} = [0.3,0.2]^T$, $d_{i,1} = 0.1\cos^2 t$, and

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(a) Tracking errors of 1-th agent. (b) Tracking errors of 2-th agent.

(c) Tracking errors of 3-th agent. (d) Tracking errors of 4-th agent.

Fig. 4. Tracking errors of the controlled MAS. (a) Tracking errors of first agent. (b) Tracking errors of second agent. (c) Tracking errors of third agent. (d) Tracking errors of fourth agent.

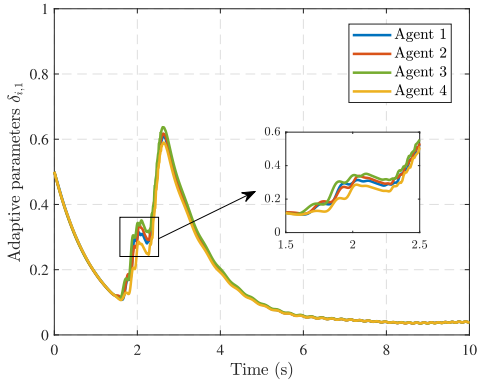


Fig. 5. Trajectories of the adaptive parameters $\hat{\delta}_{i,1}$.

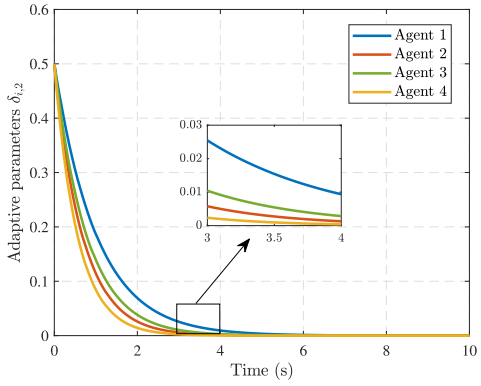


Fig. 6. Trajectories of the adaptive parameters $\hat{\delta}_{i,2}$.

$d_{i,2} = 0.1\sin^2 t$. The reference signal is selected as $y_r = 0.2\sin t + 0.5\cos t$. It is assumed that the communication channel between the reference signal and the controlled system suffers from cyber-attacks, resulting in the MAS only periodically receiving discrete signal points on the original trajectory.

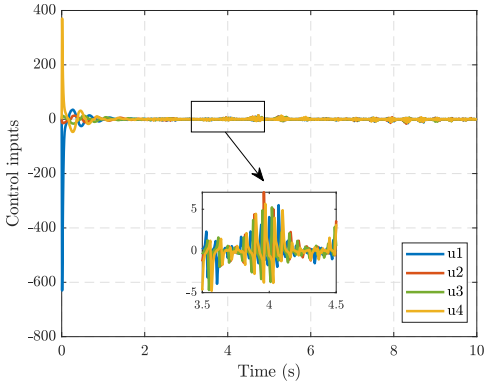


Fig. 7. Trajectories of the control inputs.

Further, we assume that the signal reception period is $2s$, and the number of signal points is 10. The initial conditions of the controlled system are selected as $x_{1,1} = 1$, $x_{2,1} = 0.7$, $x_{3,1} = 0.3$, $x_{4,1} = 0$, $x_{1,2} = 0.1$, $x_{2,2} = 0.1$, $x_{3,2} = 0.1$, $x_{4,2} = 0.1$, $\hat{\delta}_{1,1} = 0.5$, $\hat{\delta}_{1,2} = 0.5$, $\hat{\delta}_{2,1} = 0.5$, $\hat{\delta}_{2,2} = 0.5$, $\hat{\delta}_{3,1} = 0.5$, $\hat{\delta}_{3,2} = 0.5$, $\hat{\delta}_{4,1} = 0.5$, $\hat{\delta}_{4,2} = 0.5$, $\hat{y}_1 = 0.2$, $\hat{y}_2 = 0.2$, $\hat{y}_3 = 0.2$, and $\hat{y}_4 = 0.2$. The parameters of the prescribed time regulator are uniformly selected as $T_f = 3$, $\gamma = 0.9$, and $\ell = 0.1$. Meanwhile, the model of actuator fault is chosen as $u_i = (1 - 0.1\sin^2 t)\upsilon_i + 0.1\sin^2 t$. To enable a comparative assessment, we benchmark the proposed control method (44) against the prominent fixed-time control algorithm developed in [42]. (Example provides specifications for plant modeling and design parameter selection.) The communication topology of the controlled MAS is shown in Fig. 1, and the simulation results are presented in Figs. 2–7.

As shown in Fig. 2, in the comparison experiments of different trajectory reconstruction algorithms, the proposed method for reconstructing discrete signals clearly outperforms the event-triggered communication. Fig. 3 presents the outputs of the controlled system and the reconstructed trajectory, illustrating that all agents can accurately track the reconstructed reference signal. The tracking errors, shown in Fig. 4, reveal that all errors converge to the specified set within the prescribed tracking time. Figs. 5 and 6 depict the trajectories of the adaptive parameters, indicating that the compensation signals remain bounded throughout the control process. In the end, the performance of the control inputs is presented in Fig. 7, demonstrating that all the control inputs are bounded.

V. Conclusion

In this article, the prescribed-time fault-tolerant tracking control strategy has been developed for nonlinear MASs with parameter uncertainties and external disturbances. The configuration of prescribed-time regulators and self-regulating control gains for the controlled system guarantees precise tracking within the prescribed time. The trajectory reconstruction method based on cubic spline interpolation has further improved control performance while preventing transient overloads. Stability analysis and simulation results demonstrate the effectiveness and applicability of the proposed control strategy in engineering systems. These results confirm the control

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strategy's potential to enhance the robustness and efficiency of MASs under uncertain conditions.

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